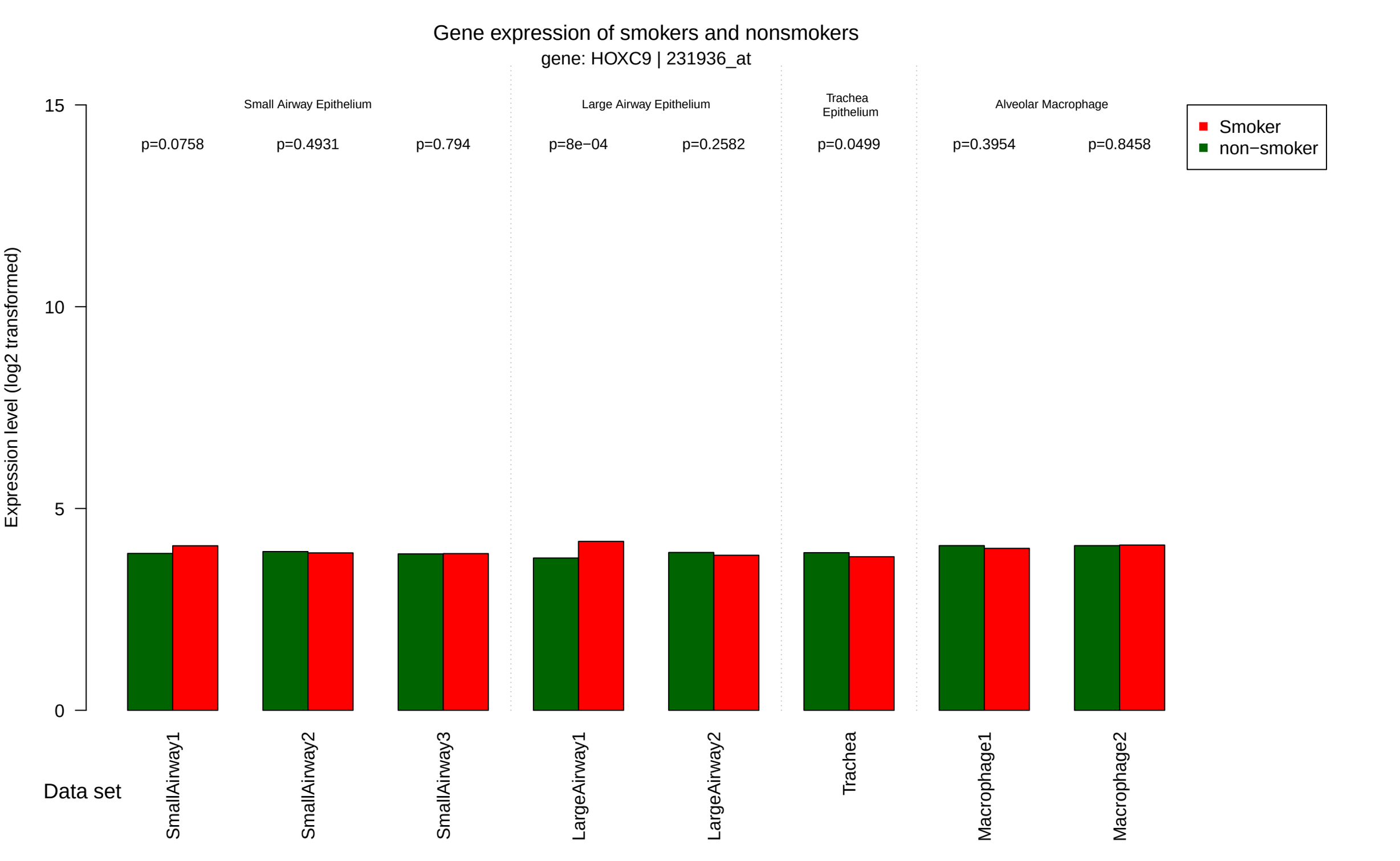
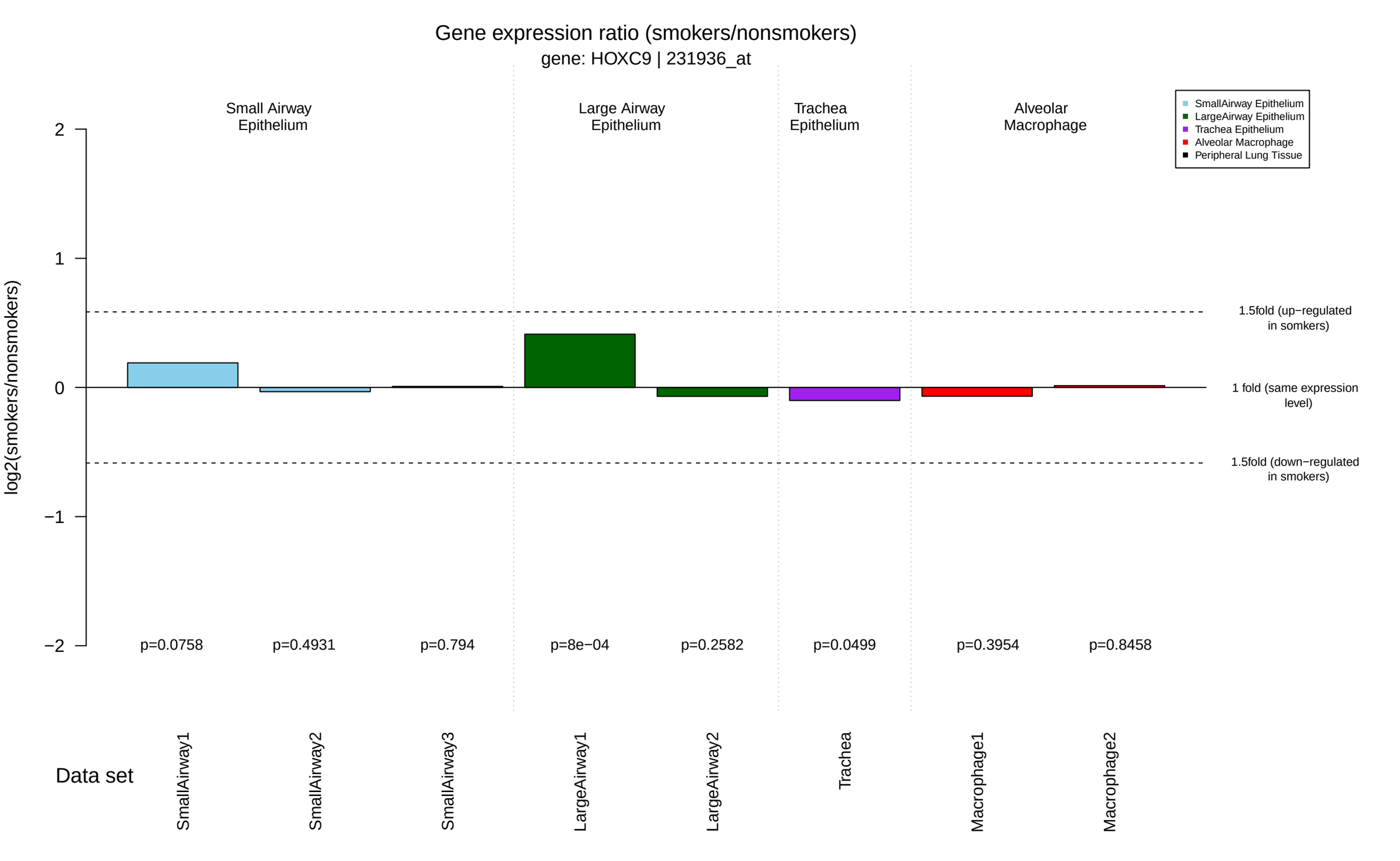
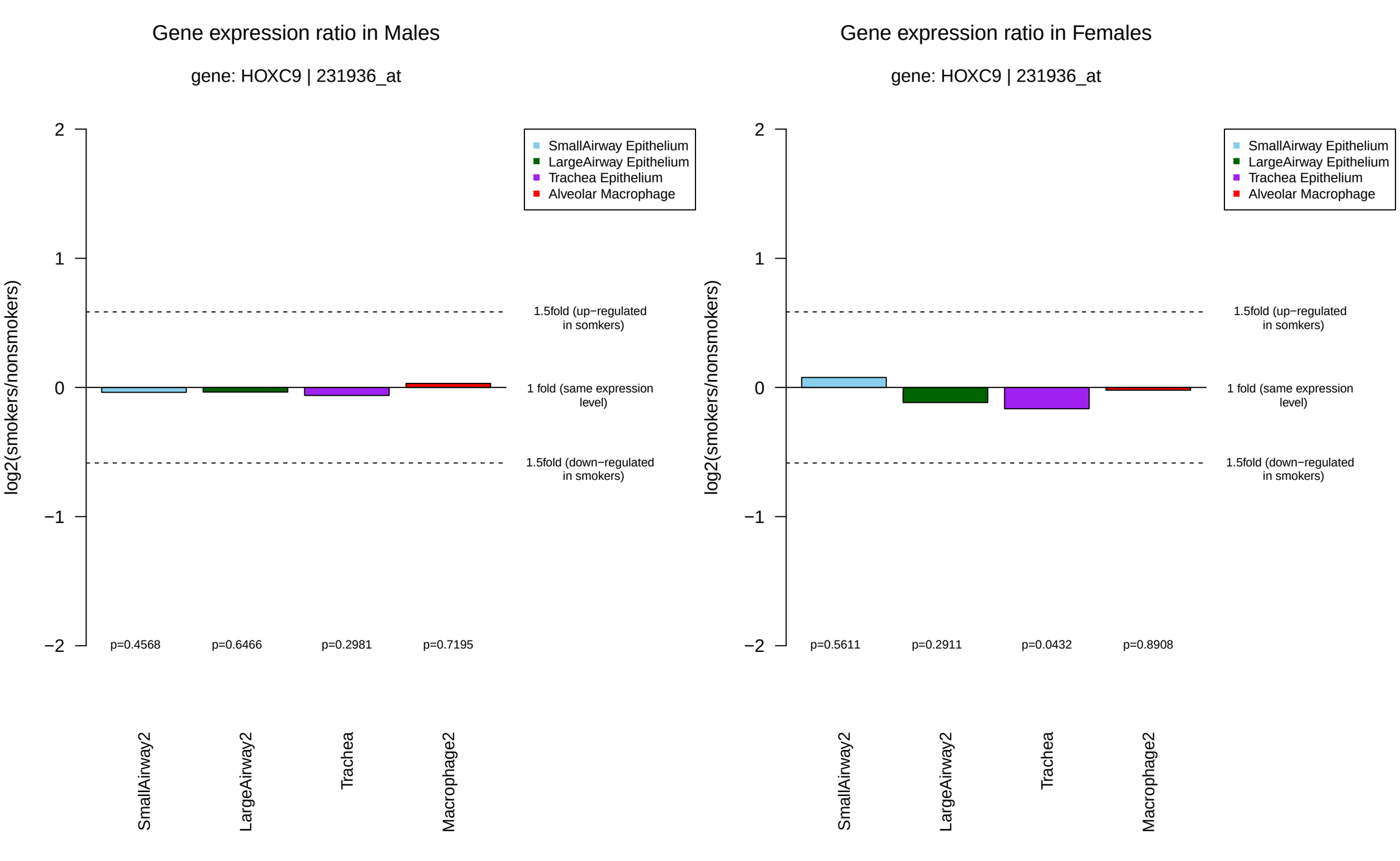
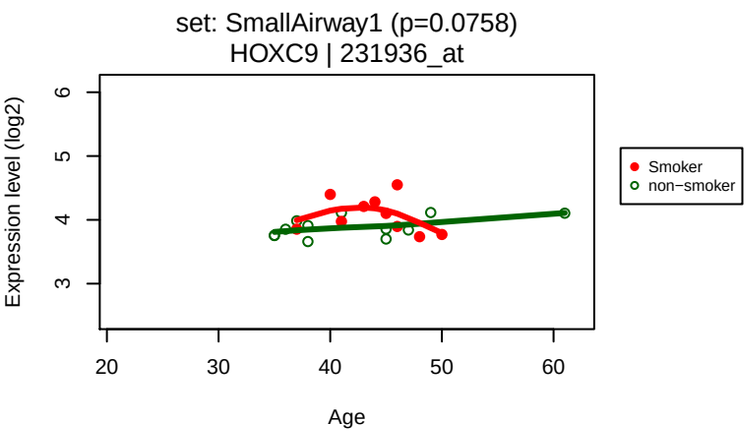
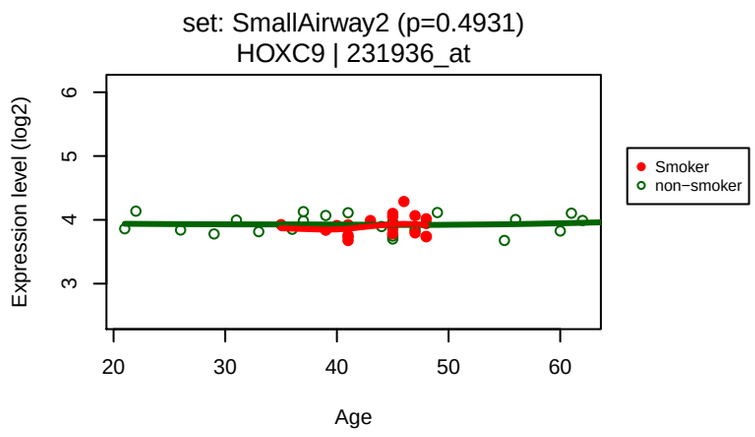
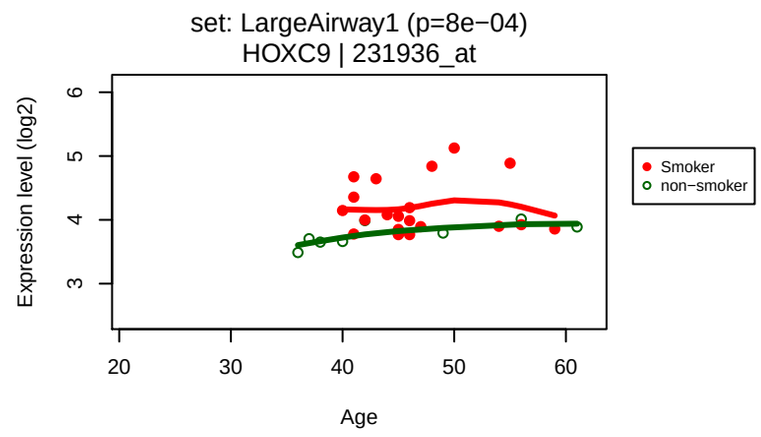

Gene expression of smokers and nonsmokers
gene: HOXC9 | 231936_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0758
p=0.4931
p=0.794
p=8e−04
p=0.2582
p=0.0499
p=0.3954
p=0.8458
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

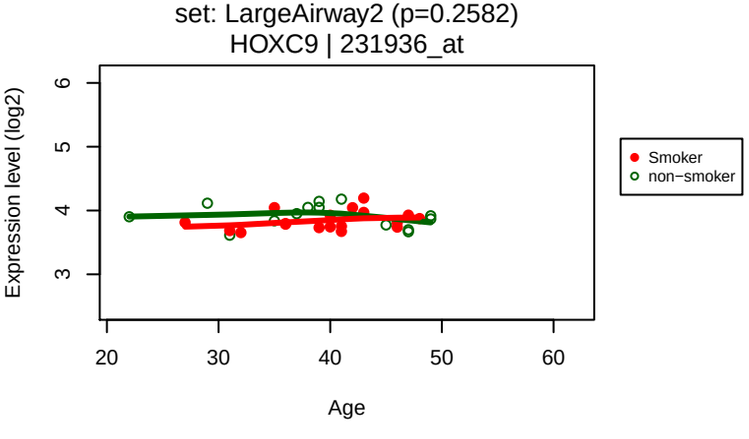

Gene expression ratio (smokers/nonsmokers)
gene: HOXC9 | 231936_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0758
p=0.4931
p=0.794
p=8e−04
p=0.2582
p=0.0499
p=0.3954
p=0.8458
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

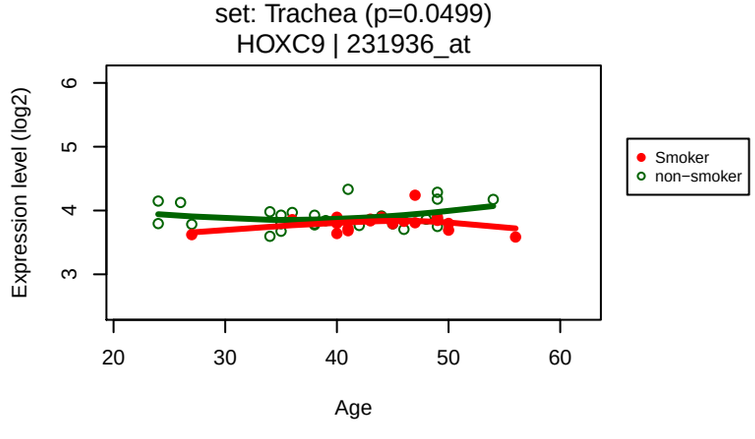

Gene expression ratio in Males
gene: HOXC9 | 231936_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4568
p=0.6466
p=0.2981
p=0.7195
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXC9 | 231936_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5611
p=0.2911
p=0.0432
p=0.8908
SmallAirway2
LargeAirway2
Trachea
Macrophage2

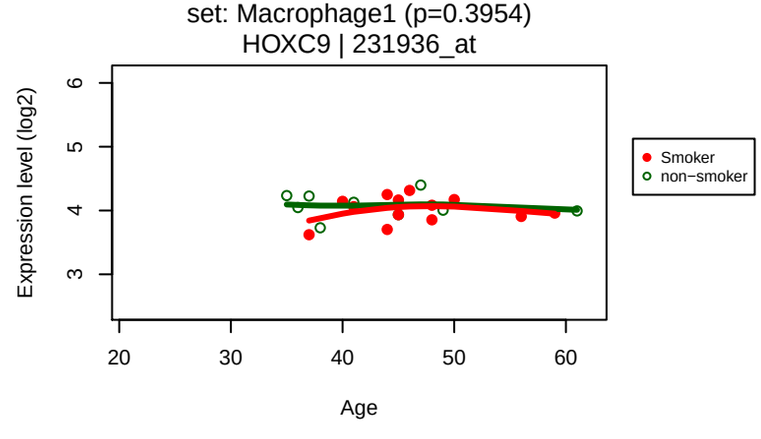

set: SmallAirway1 (p=0.0758)
HOXC9 | 231936_at
Expression level (log2)
Age
Smoker
non−smoker

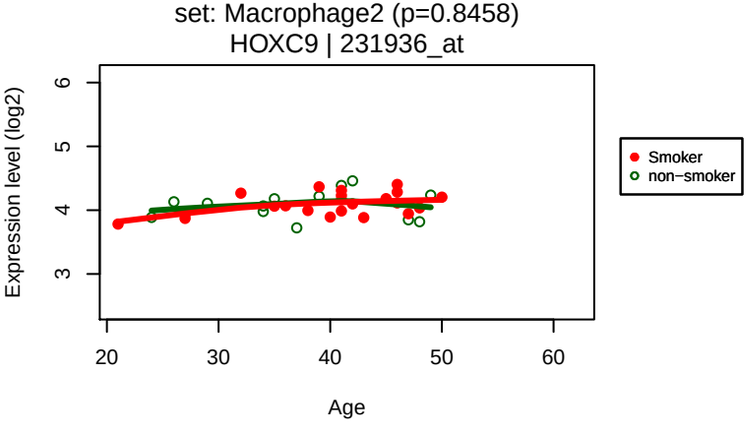

set: SmallAirway2 (p=0.4931)
HOXC9 | 231936_at
Expression level (log2)
Age
Smoker
non−smoker

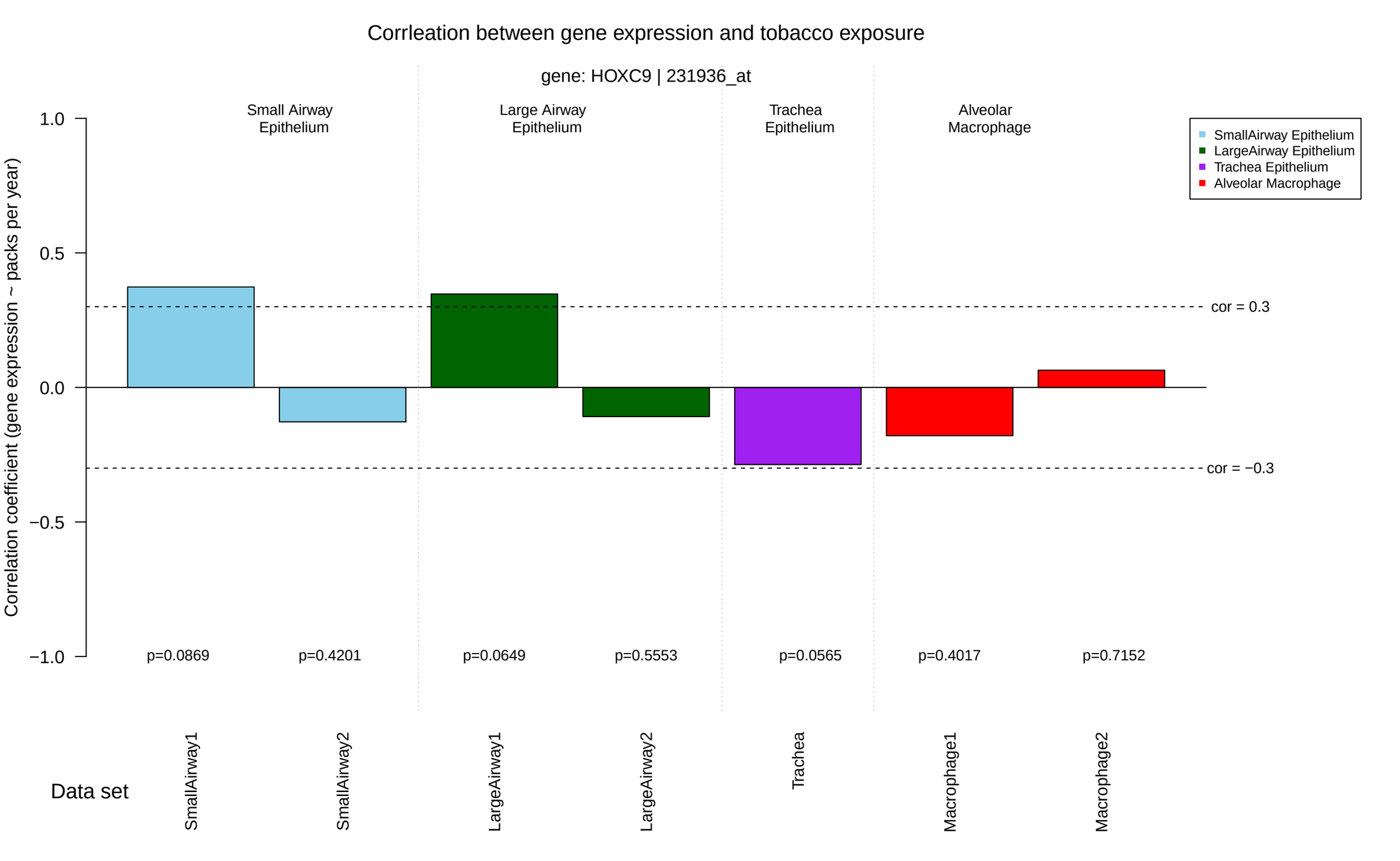

set: LargeAirway1 (p=8e−04)
HOXC9 | 231936_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2582)
HOXC9 | 231936_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0499)
HOXC9 | 231936_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3954)
HOXC9 | 231936_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8458)
HOXC9 | 231936_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXC9 | 231936_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0869
p=0.4201
p=0.0649
p=0.5553
p=0.0565
p=0.4017
p=0.7152
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2